Chapter 9: Operations on Integers

Worksheet 9: Operations on Integers

Learning Objective: Perform addition, subtraction, multiplication and division of integers, and apply the properties of operations in real contexts such as temperature, elevation and money.

Worked Example:
The temperature at night was $-4°C$ and it dropped further by $6°C$. What was the new temperature?

Solution: $-4 + (-6) = -10°C$.

Questions (1–20)

1. Add: $(-8) + 5$.

2. Subtract: $6 - (-3)$.

3. Multiply: $(-7) \times 4$.

4. Divide: $(-36) \div (-6)$.

5. What is the additive inverse of -15?

6. A submarine is at -250 m (250 m below sea level). It rises by 90 m. What is its new position?

7. The temperature was 5°C and dropped by 12°C. Find the new temperature.

8. Multiply: $(-9) \times (-6)$.

9. A man had a debt of Rs. 500 (-500). He earned Rs. 800. What is his balance now?

10. Evaluate: $(-12) + 7 - (-5)$.

11. Evaluate using the distributive property: $(-6) \times (15 + 5)$.

12. A lift goes down 5 floors from floor 12, then up 8 floors. At which floor does it end up?

13. Find the product: $(-4) \times 3 \times (-2)$.

14. The temperature in a city changed from -6°C to 9°C over a day. By how many degrees did it rise?

15. Divide: $84 \div (-7)$.

16. A scuba diver descends 18 m, then rises 7 m, then descends another 12 m. What is the diver's final position relative to the surface?

17. Verify the associative property of addition using -5, 8 and -3: check if $(-5+8)+(-3) = -5+(8+(-3))$.

18. A company's profit or loss over 4 months was $-$Rs. 2,000, +Rs. 5,500, $-$Rs. 1,200, +Rs. 3,700. Find the net profit or loss.

19. Find today's temperature where you live and a typical winter temperature in a cold hill station (which can be negative). Find the difference between the two.

20. If you spend Rs. 150 and then receive Rs. 500 as pocket money, express the net change to your money using integers.

Reflection Question: Where in real life have you seen or used negative numbers, apart from temperature? Write one example.

Real-Life Application Question: Check a weather report or app for today's temperature in a hill station or cold region. Note whether it is positive or negative, and explain what a negative temperature means.